

Challenge 20: Torsional levitated optomechanics

Derivation of the librational frequency ω_t and of the light-induced torsion-torsion coupling g

Companion implementation: `challenges/20/answer.py`

1 Statement and strategy

Two identical dielectric prolate ellipsoids (semi-major axis a , two equal semi-minor axes b , relative permittivity ϵ_r , mass density ρ) are levitated in two Gaussian optical tweezers that propagate along $\hat{z}$, are linearly polarised along $\hat{x}$, carry power P_0 , have waist w_0 and wave vector k , and whose foci are separated by R along $\hat{x}$. The long axes align with the polarisation and perform small librations about it. We must produce ω_t and g in

$$H = \hbar\omega_t (a_1^\dagger a_1 + a_2^\dagger a_2) + \hbar g (a_1^\dagger a_2 + a_1 a_2^\dagger). \quad (1)$$

The derivation has four steps: (i) the quasi-static polarisability tensor of a sub-wavelength ellipsoid; (ii) the orientational potential of one particle in its tweezer, giving ω_t ; (iii) the light-induced (optical-binding) dipole-dipole energy expanded to bilinear order in the two libration angles; (iv) canonical quantisation plus the rotating-wave approximation, giving g .

2 Polarisability of the ellipsoid

For $a, b \ll \lambda$ the particle responds as a point dipole with the quasi-static polarisability tensor $\alpha = \text{diag}(\alpha_\parallel, \alpha_\perp, \alpha_\perp)$ in the body frame [8, 9, 10],

$$\alpha_j = \frac{\epsilon_0 V (\epsilon_r - 1)}{1 + L_j (\epsilon_r - 1)}, \quad V = \frac{4\pi}{3} ab^2, \quad (2)$$

with depolarisation factors, for eccentricity $e = \sqrt{1 - b^2/a^2}$,

$$L_\parallel = \frac{1 - e^2}{e^3} \left[\frac{1}{2} \ln \frac{1 + e}{1 - e} - e \right], \quad L_\perp = \frac{1 - L_\parallel}{2}. \quad (3)$$

The single quantity that controls both answers is the polarisability anisotropy

$$\Delta\alpha \equiv \alpha_\parallel - \alpha_\perp = \epsilon_0 V (\epsilon_r - 1)^2 \frac{L_\perp - L_\parallel}{[1 + L_\parallel (\epsilon_r - 1)][1 + L_\perp (\epsilon_r - 1)]} > 0. \quad (4)$$

3 Librational frequency ω_t

Let $\hat{n}$ be the long axis and θ the angle between $\hat{n}$ and the polarisation $\hat{x}$. Decomposing $\mathbf{E} = E_0 \hat{x}$ along the principal axes, the cycle-averaged interaction energy of the induced dipole is

$$U(\theta) = -\frac{1}{4} E_0^2 [\alpha_\parallel \cos^2 \theta + \alpha_\perp \sin^2 \theta] = \text{const} + \frac{1}{4} \Delta\alpha E_0^2 \sin^2 \theta, \quad (5)$$

which is Eq. (1) of Hoang *et al.* [3] written with $\tilde{\chi}_j = \alpha_j/(\epsilon_0 V)$. Expanding to $\mathcal{O}(\theta^2)$ gives the angular spring constant

$$k_{\text{tor}} = \left. \frac{\partial^2 U}{\partial \theta^2} \right|_0 = \frac{1}{2} \Delta\alpha E_0^2 = \frac{\Delta\alpha I_0}{\epsilon_0 c}, \quad I_0 = \frac{2P_0}{\pi w_0^2}, \quad E_0^2 = \frac{2I_0}{\epsilon_0 c} = \frac{4P_0}{\pi w_0^2 \epsilon_0 c}. \quad (6)$$

Here I_0 is the peak intensity of the Gaussian tweezer and E_0 the field *amplitude* at the focus. Because the spheroid is symmetric about $\hat{n}$, the two librational planes containing $\hat{x}$ are degenerate and rotation about $\hat{n}$ is free [11]: there is exactly one torsional mode per particle, as assumed in Eq. (1). With mass $m = \rho V$ and the transverse centroidal moment of inertia

$$I = \frac{m(a^2 + b^2)}{5} = \frac{4\pi}{15} \rho ab^2 (a^2 + b^2), \quad (7)$$

the librational frequency is $\omega_t = \sqrt{k_{\text{tor}}/I}$, i.e.

Result 1: librational frequency

$$\omega_t = \sqrt{\frac{\Delta\alpha I_0}{\epsilon_0 c I}} = \sqrt{\frac{2 \Delta\alpha P_0}{\pi \epsilon_0 c w_0^2 I}} = \sqrt{\frac{15 \Delta\alpha P_0}{2\pi^2 \epsilon_0 c \rho w_0^2 ab^2 (a^2 + b^2)}} \quad (8)$$

Written with the normalised anisotropy $\Delta\tilde{\chi} = \Delta\alpha/(\epsilon_0 V)$ this is $\omega_t = [10 \Delta\tilde{\chi} P_0/(\pi \rho c w_0^2 (a^2 + b^2))]^{1/2}$, algebraically identical to Eq. (2) of Ref. [3], which was verified experimentally through its $\omega_t \propto \sqrt{P_0}$ scaling.

4 Light-induced dipole–dipole coupling

4.1 Tilt-induced transverse dipoles

For a tilt $\theta_i \ll 1$ of particle i in a plane containing $\hat{x}$ (unit vector $\hat{e}_\perp \perp \hat{x}$ in that plane; $\hat{e}_\perp = \hat{y}$ or $\hat{z}$ — both give the same result, see Sec. 5), the induced dipole is

$$\mathbf{p}_i = \alpha_\perp E_0 \hat{x} + \Delta\alpha E_0 \cos \theta_i \hat{n}_i = E_0 [(\alpha_\parallel - \Delta\alpha \theta_i^2) \hat{x} + \Delta\alpha \theta_i \hat{e}_\perp] + \mathcal{O}(\theta_i^3). \quad (9)$$

4.2 Interaction energy

Two oscillating dipoles separated by $\mathbf{R} = R\hat{x}$ interact through the retarded dipole field. Writing the instantaneous energy $-\mathbf{p}_1 \cdot \mathbf{E}_2(\mathbf{r}_1)$ with the free-space dyadic Green function [7, 4, 5],

$$U_{\text{int}} = C \operatorname{Re} \left\{ \frac{e^{ikR}}{4\pi\epsilon_0 R^3} \left[(\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{R})(\mathbf{p}_2 \cdot \hat{R}))(1 - ikR) - k^2 R^2 (\mathbf{p}_1 \times \hat{R}) \cdot (\mathbf{p}_2 \times \hat{R}) \right] \right\}. \quad (10)$$

The prefactor C is discussed in the boxed remark below: $C = \frac{1}{2}$ for the correct cycle average, $C = 1$ in the convention of Ref. [1].

Insert Eq. (9) with $\hat{R} = \hat{x}$ and collect the terms bilinear in $\theta_1 \theta_2$. Since the transverse components are perpendicular to $\hat{R}$, they do not appear in $(\mathbf{p}_i \cdot \hat{R})$, so the middle term contributes nothing; the remaining two give $\mathbf{p}_1 \cdot \mathbf{p}_2 \supset \Delta\alpha^2 E_0^2 \theta_1 \theta_2$ and $(\mathbf{p}_1 \times \hat{R}) \cdot (\mathbf{p}_2 \times \hat{R}) \supset \Delta\alpha^2 E_0^2 \theta_1 \theta_2$. Hence

$$U_{\text{int}}^{(\theta_1 \theta_2)} = C \Delta\alpha^2 E_0^2 \theta_1 \theta_2 \operatorname{Re} \left[\frac{e^{ikR} (1 - ikR - k^2 R^2)}{4\pi\epsilon_0 R^3} \right] = -C \Delta\alpha^2 E_0^2 \operatorname{Re} G_\perp \theta_1 \theta_2, \quad (11)$$

where the transverse component of the Green function and its real part are

$$G_\perp = \frac{e^{ikR} (k^2 R^2 + ikR - 1)}{4\pi\epsilon_0 R^3}, \quad \operatorname{Re} G_\perp = \frac{(k^2 R^2 - 1) \cos kR - kR \sin kR}{4\pi\epsilon_0 R^3}. \quad (12)$$

Note that $\operatorname{Re} G_\perp$ decays only as $1/R$ in the far field, so the torsion–torsion interaction is long-ranged and oscillates in sign with kR .

4.3 Quantisation

The two librators plus Eq. (11) give

$$H = \sum_{i=1,2} \left[\frac{L_i^2}{2I} + \frac{1}{2} k_{\text{tor}} \theta_i^2 \right] + \kappa \theta_1 \theta_2, \quad \kappa = -C \Delta \alpha^2 E_0^2 \text{Re } G_{\perp}. \quad (13)$$

With $\theta_i = \theta_{\text{zpf}}(a_i + a_i^\dagger)$ and $\theta_{\text{zpf}} = \sqrt{\hbar/2I\omega_t}$, and dropping the counter-rotating $a_1 a_2 + \text{h.c.}$ terms (valid since $|g| \ll \omega_t$), one obtains Eq. (1) with $\hbar g = \kappa \theta_{\text{zpf}}^2$:

Result 2: torsion–torsion coupling

$$g = -C \frac{\Delta \alpha^2 E_0^2 \text{Re } G_{\perp}}{2I\omega_t} = -C \frac{\Delta \alpha^2 P_0 [(k^2 R^2 - 1) \cos kR - kR \sin kR]}{2\pi^2 \epsilon_0^2 c w_0^2 R^3 I \omega_t} \quad (14)$$

with I from Eq. (7), ω_t from Eq. (8), $E_0^2 = 4P_0/(\pi w_0^2 \epsilon_0 c)$, and $C = \frac{1}{2}$ ($C = 1$ in the convention of Ref. [1]).

Because $\Delta \alpha^2 E_0^2/(2I) = k_{\text{tor}} \Delta \alpha = \omega_t^2 I \Delta \alpha$, Eq. (14) collapses to the compact, power-independent ratio

$$\frac{g}{\omega_t} = -C \Delta \alpha \text{Re } G_{\perp}, \quad (15)$$

a useful check: both g and ω_t scale as $\sqrt{P_0}$, so their ratio is a pure function of geometry and kR . The normal modes of Eq. (1) are the symmetric/antisymmetric librations at $\omega_{\pm} = \omega_t \pm g$.

FLAG — the factor of two in g

Equation (10) with $C = 1$ is the *instantaneous* electrostatic-like energy $U = -\mathbf{p}_1 \cdot \mathbf{E}_2$ evaluated with field *amplitudes*. The dipoles oscillate at the optical frequency, so the physical interaction is the cycle average, which carries an extra factor $\frac{1}{2}$: $C = \frac{1}{2}$. This is the convention of the supplementary material of Rieser *et al.* [4], whose optical-binding force $\mathbf{F}_{\text{bind}} = \alpha_j \nabla_j \text{Re}[(\alpha_{j'}/2\epsilon_0) \mathbf{E}_0^*(\mathbf{r}_j) \cdot \mathbf{G} \cdot \mathbf{E}_0(\mathbf{r}_{j'})]$ is exactly the gradient of $-\frac{1}{2\epsilon_0} \text{Re}[\mathbf{p}_1^* \cdot \mathbf{G} \cdot \mathbf{p}_2]$, and of the quantum treatment of Rudolph *et al.* [5, 6].

Reference [1], from which this problem is taken, uses $C = 1$ in its Eqs. (4)–(9) while it *does* cycle-average the trapping term ($k_{\text{tor}} = \Delta \alpha E_0^2/2$, our Eq. (6)). Its g_0 is therefore a factor of two larger than the cycle-averaged result. Both are implemented in `answer.py` via the `convention` keyword; the cross-check of Sec. 7 is quoted with $C = 1$ so that it compares like with like. ω_t is unaffected by this ambiguity.

5 Remarks on the geometry and on neglected terms

Polarisation parallel vs. perpendicular to $\mathbf{R}$. The problem places both the polarisation and the separation along $\hat{x}$ (head-to-tail dipoles), whereas Ref. [1] polarises along $\hat{y}$ with the separation along $\hat{x}$ (side-by-side dipoles). This does *not* change g : in both cases the tilt-induced transverse dipole components are perpendicular to $\hat{R}$, so they drop out of $(\mathbf{p}_i \cdot \hat{R})$ and the bilinear term of Eq. (11) is the same. Only the θ_i^2 terms differ between the two geometries.

Binding-induced frequency shift. The $\mathcal{O}(\theta_i^2)$ part of Eq. (10) renormalises k_{tor} . In the head-to-tail geometry of this problem the θ_i^2 terms of $\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{R})(\mathbf{p}_2 \cdot \hat{R})$ give a relative correction $4\alpha_{\parallel} [\cos kR + kR \sin kR]/(4\pi\epsilon_0 R^3) \approx 2\%$ to ω_t^2 (i.e. $\sim 1\%$ to ω_t) at $R \simeq \lambda$ for the parameters of Table 1. It is neglected in Eq. (8), consistent with Ref. [1].

Beyond the dipole approximation. For the benchmark parameters $a/\lambda = 0.28$, so quadrupole and higher multipoles are not negligible; Appendix A of Ref. [1] estimates corrections of a few per-cent. Likewise, optical binding is genuinely non-reciprocal and non-Hermitian [4, 5]; the Hermitian beam-splitter form assumed in Eq. (1) is the conservative part only.

6 Numerical evaluation

Table 1 evaluates the results for the parameters of Ref. [1] ($a = 300$ nm, $b = 180$ nm, $\epsilon_r = 5.7$, $\rho = 3500$ kg m⁻³, $w_0 = 500$ nm, $\lambda = 1064$ nm, $R = 1.06$ μ m).

Quantity	This work	Ref. [1]
V	4.0715×10^{-20} m ³	0.04×10^{-18} m ³
$L_{\parallel}$	0.20996	0.210
$L_{\perp}$	0.39502	0.395
$\Delta\alpha$	2.5966×10^{-31} C m ² V ⁻¹	—
I	3.4885×10^{-30} kg m ²	3.49×10^{-30} kg m ²
$\omega_t/2\pi$ (600 mW)	1.0418 MHz	—
$\omega_t/2\pi$ (1000 mW)	1.3449 MHz	—
$ g /2\pi$ (1000 mW, $C = \frac{1}{2}$)	50.5 kHz	—
$ g /2\pi$ (1000 mW, $C = 1$)	101.0 kHz	≈ 119 kHz (quoted)
$ g /\omega_t$ ($C = 1$)	0.0751	—

Table 1: Derived quantities for the benchmark parameters.

7 Cross-check against the published data

The authors of Ref. [1] deposited the data behind their Fig. 2(b) — $g_0/2\pi$ versus R/λ at three tweezer powers — on Zenodo [2]. Table 2 compares those numbers with Eq. (14) evaluated with $C = 1$ (their convention).

R/λ	600 mW		800 mW		1000 mW		ratio
	Zenodo	Eq. (14)	Zenodo	Eq. (14)	Zenodo	Eq. (14)	
0.550	-151.24	+113.29	-174.63	+130.81	-195.24	+146.25	-1.3350
0.700	-11.67	+8.74	-13.48	+10.10	-15.07	+11.29	-1.3350
0.850	+89.93	-67.37	+103.85	-77.79	+116.10	-86.97	-1.3350
0.950	+109.29	-81.87	+126.20	-94.54	+141.10	-105.69	-1.3350
0.996	+104.46	-78.25	+120.63	-90.36	+134.86	-101.02	-1.3350
1.200	+15.73	-11.78	+18.16	-13.60	+20.31	-15.21	-1.3350
1.500	-70.11	+52.52	-80.96	+60.65	-90.52	+67.80	-1.3350
2.000	+52.85	-39.59	+61.02	-45.71	+68.23	-51.11	-1.3350
3.000	+35.36	-26.48	+40.83	-30.58	+45.64	-34.19	-1.3350
5.000	+21.25	-15.92	+24.54	-18.38	+27.44	-20.55	-1.3350

Table 2: $g/2\pi$ in kHz. “Zenodo” is the tabulated Fig. 2(b) data of Ref. [2]; “Eq. (14)” is this derivation with $C = 1$. The last column is Zenodo/ours at 1000 mW. Generated by `answer.py`.

Three observations:

1. The R **dependence is reproduced exactly**. Over $0.55 \leq R/\lambda \leq 6.0$ (2725 tabulated points, of which ten are shown) the ratio is constant to 15 significant digits, so Eqs. (12)–(14) carry the correct retarded Green function, including the sign oscillations and the $1/R$ far-field tail.

2. The **power dependence is reproduced exactly**: the tabulated values scale as $\sqrt{P_0}$ (e.g. $109.29/141.10 = 0.7745 = \sqrt{0.6}$), confirming that the $1/\omega_t$ of Eq. (14) — and not $g \propto E_0^2 \propto P_0$ as stated in the text of Ref. [1] — is what their code implements.
3. Two constant discrepancies remain. The **overall sign** is a convention: Ref. [1] writes $g_0 = +\Delta\alpha^2 E_0^2 \text{Re } G_\perp / (2\sqrt{I_1 I_2 \omega_1 \omega_2})$, whereas Eq. (1) with the potential (11) fixes the opposite sign. The residual **factor** 1.334977 is not accounted for by anything in the published equations: it is not the multipole correction of their Appendix A (their own $C_1 + C_2$ gives only 1.045), not rounding of $L_\parallel, L_\perp$ (-0.06%), and not the choice of c (-0.03%); ϵ_0 cancels from g identically. It appears to be an undocumented constant inside their plotting code. Their quoted 119 kHz lies between our 101 kHz and their own curve’s 135 kHz at the same R and $P_0 = 1$ W.

Independently, ω_t of Eq. (8) is algebraically identical to the experimentally validated expression of Hoang *et al.* [3], which is a second, independent confirmation of that half of the answer.

References

- [1] G. Zhang, H. Zhang and Z. Yin, *Scalable universal quantum gates between nitrogen-vacancy centers in levitated nanodiamond arrays*, Phys. Rev. A **112**, 032611 (2025), [arXiv:2504.08194](#).
- [2] G. Zhang, H. Zhang and Z. Yin, *Quantum gates between nitrogen-vacancy centers in levitated nanodiamond arrays* (dataset), [10.5281/zenodo.16917848](#) (2025).
- [3] T. M. Hoang, Y. Ma, J. Ahn, J. Bang, F. Robicheaux, Z.-Q. Yin and T. Li, *Torsional optomechanics of a levitated nonspherical nanoparticle*, Phys. Rev. Lett. **117**, 123604 (2016).
- [4] J. Rieser *et al.*, *Tunable light-induced dipole-dipole interaction between optically levitated nanoparticles*, Science **377**, 987 (2022).
- [5] H. Rudolph, U. Delić, K. Hornberger and B. A. Stickler, *Quantum theory of non-Hermitian optical binding between nanoparticles*, Phys. Rev. A **110**, 063507 (2024).
- [6] H. Rudolph, U. Delić, K. Hornberger and B. A. Stickler, *Quantum optical binding of nanoscale particles*, Phys. Rev. Lett. **133**, 233603 (2024).
- [7] K. Dholakia and P. Zemanek, *Colloquium: Grippled by light — optical binding*, Rev. Mod. Phys. **82**, 1767 (2010).
- [8] L. D. Landau and E. M. Lifshitz, *Electrodynamics of Continuous Media*, 2nd ed., Sec. 4.
- [9] J. A. Osborn, *Demagnetizing factors of the general ellipsoid*, Phys. Rev. **67**, 351 (1945).
- [10] C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles*, Ch. 5 (Wiley, 1983).
- [11] B. A. Stickler, K. Hornberger and M. S. Kim, *Quantum rotations of nanoparticles*, Nat. Rev. Phys. **3**, 589 (2021).